Formula 1 Podium Prediction

Milestone 1: Analytical Report

Predicting, for every driver in a race, whether they finish in the top three, using only information available after qualifying and before the race starts.

Dataset: Formula 1 World Championship (1950–2024), Kaggle / Ergast.

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Part 1: Data Cleaning and Auditing

F1 Podium Prediction, Milestone 1

1. Scope and approach

We converted the 14 raw Ergast/Kaggle tables into a validated modelling table where each row is one driver in one race. The target is `podium` (best `positionOrder` ≤ 3). The prediction cutoff is after qualifying and before the race. The raw CSVs were never modified, and all cleaning ran on typed copies.

2. Initial audit (before cleaning)

- **Placeholders:** 30 of 120 columns contain placeholders. `\N` is the main sentinel. The only other one is the empty string (68 values, all in `qualifying.q2/q3`). `positionOrder`, the target source, has none. *(Fig. 1)*

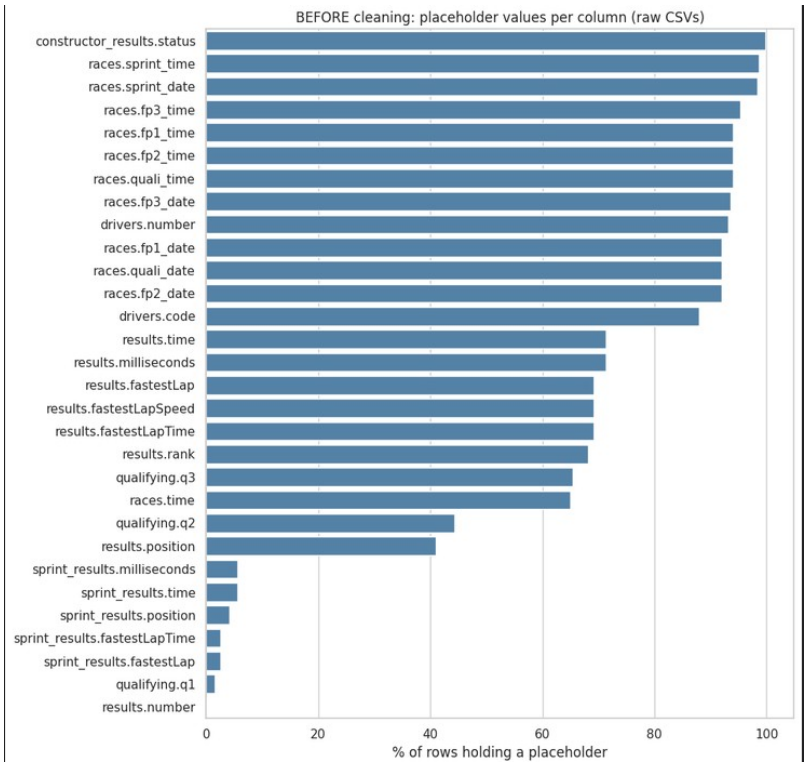


Figure 1. Share of rows holding a placeholder, per column (raw CSVs).

- **Hidden sentinel:** `grid = 0` appears in 1,638 rows (6.1%), peaking in the 1980s at 12.7%. A model would read it as "started ahead of pole", so we treat it as missing.
- **Keys:** all 14 primary keys are unique and non-null, and all 23 foreign keys reconcile with no orphans. Constructor 88 (Eagle) has no results and is unused.

3. Cleaning and typing

`\N` and empty strings became NaN. We applied nullable integers, floats, dates, timedeltas and lap times in seconds. There were 0 failed conversions across 94 columns. Parsed seconds match the

`milliseconds` columns exactly for all 589,081 lap times and 11,371 pit stops. After cleaning, 0 placeholder strings remain, every column's NaN count equals its raw placeholder count, and all keys still hold. *(Fig. 2)*

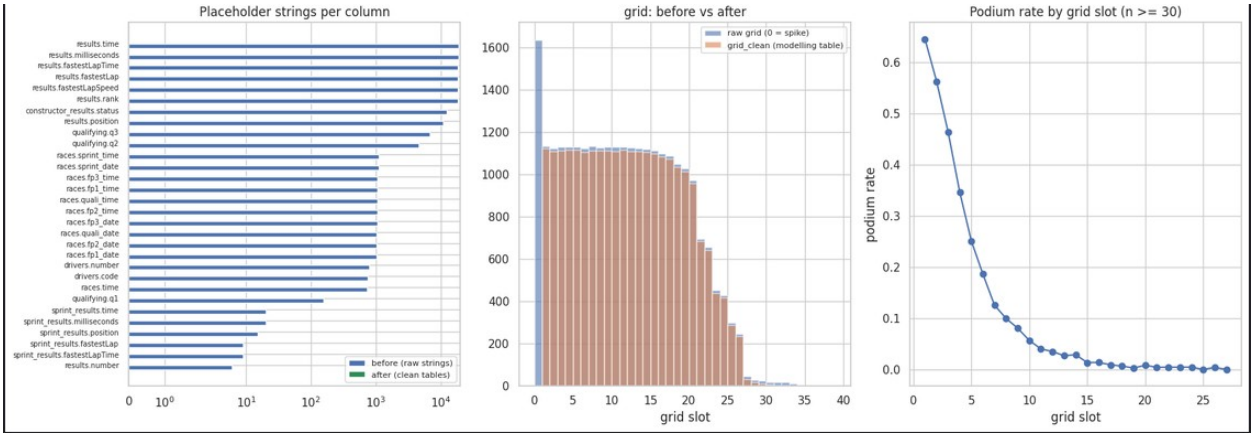


Figure 2. Left: placeholder strings before and after cleaning. Middle: grid before and after (the spike at 0 is removed). Right: podium rate by grid slot.

Outliers: one qualifying Q1 time (Blundell, 1995 Japanese GP, 16:42.640) is 9.9× its race median, while the next-worst value is 1.43×. It was set to NaN and flagged, not guessed. The 708 extreme lap times and 481 long pit stops are real red-flag stoppages. They were flagged and kept, and they are post-race tables that never become features. The fastest times all come from the short Sakhir 2020 layout.

4. Fixing the grain: shared drives

85 (raceId, driverId) pairs were duplicated, which is 91 surplus rows, mostly in 1950-58. The glossary's "keep the row with a grid slot" rule resolved none of them. Comparing four candidate rules led to a split design:

- **Target:** the driver's best `positionOrder` across his rows, which matches how the sport credits shared drives.
- **Features:** `grid` and `constructorId` come from the row with the lowest positive grid. This choice does not depend on the race outcome, so it avoids tying grid to the label.

The two sources differ in 17 pairs. Eighteen races from 1950-56 now have more than three podium rows, which is correct under shared-drive credit.

5. Population rules

Step	Rows	Podium rate
Raw results	26,759	12.69%
After deduplication	26,668	12.73%
Minus Indianapolis 500 (1950-60)	26,287	12.78%
Minus non-starters	**24,724**	**13.59%**

Indianapolis 500 (381 rows, 11 races) was excluded because it has a different grid structure and its drivers rarely raced elsewhere. The 1,563 non-starters (no grid slot and zero laps, mostly DNQ, DNPQ and Withdrew) contain no podiums, so removing them cannot change a label. The 72 genuine pit-lane starts stay in, with `grid_clean = NaN` and `grid_is_zero = 1`. The final table has 1,114 races. *(Fig. 3)*

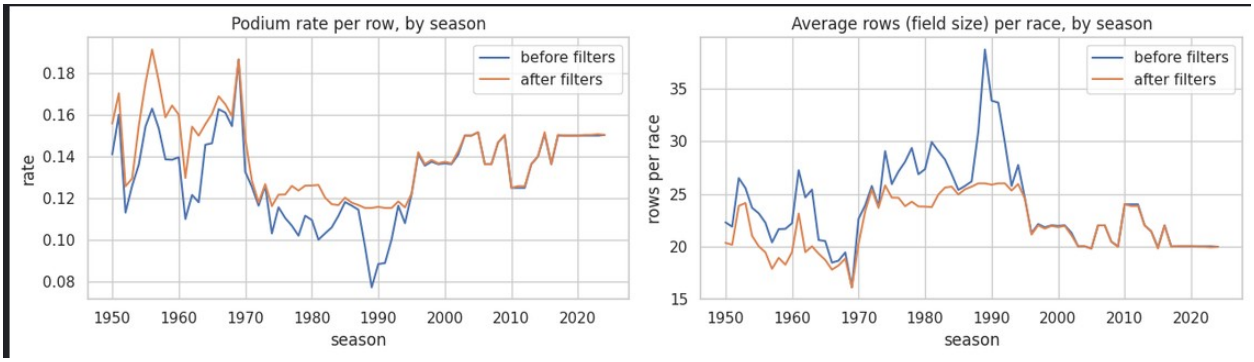


Figure 3. Podium rate per row (left) and average field size per race (right), by season, before and after the row filters.

6. Qualifying coverage and qualifying vs grid

- **Coverage drift:** qualifying data starts in 1994, drops to 6-59% of races in 1996-2002, and is sustained at ≥95% only from **2003**. Q2 starts in 2005 and Q3 in 2006. From 2006, 27% of rows have no Q2 time and 54% no Q3 time. This missingness is structural (the driver was eliminated), so it must not be mean-imputed. *(Fig. 4)*

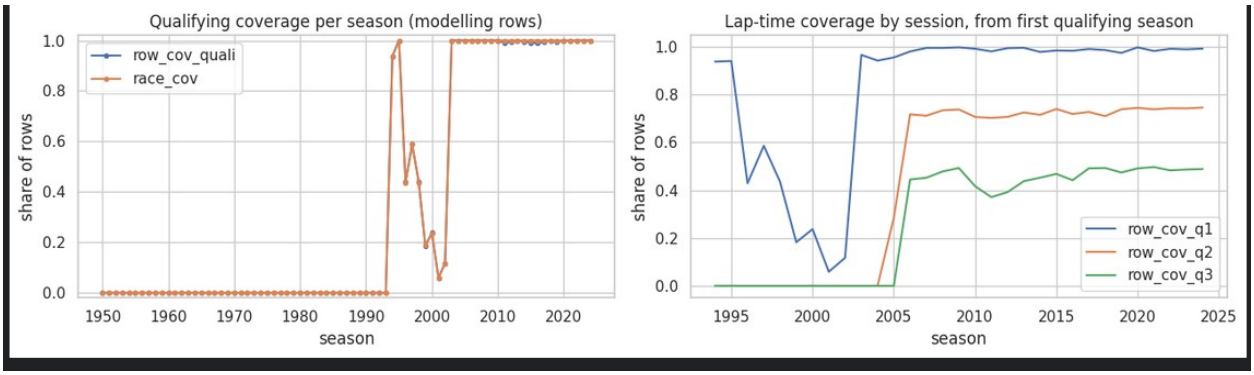


Figure 4. Qualifying coverage per season (left) and Q1/Q2/Q3 lap-time coverage (right).

- **Qualifying is not grid:** the two differ in 23.3% of rows. In 63% of those the shift is exactly one place, which is the knock-on effect of a single penalty. In 2021-22 the sprint result set the Sunday grid, and those weekends differ in 81-88% of rows against 26-29% otherwise. The effect disappears from 2023. *(Fig. 5)*

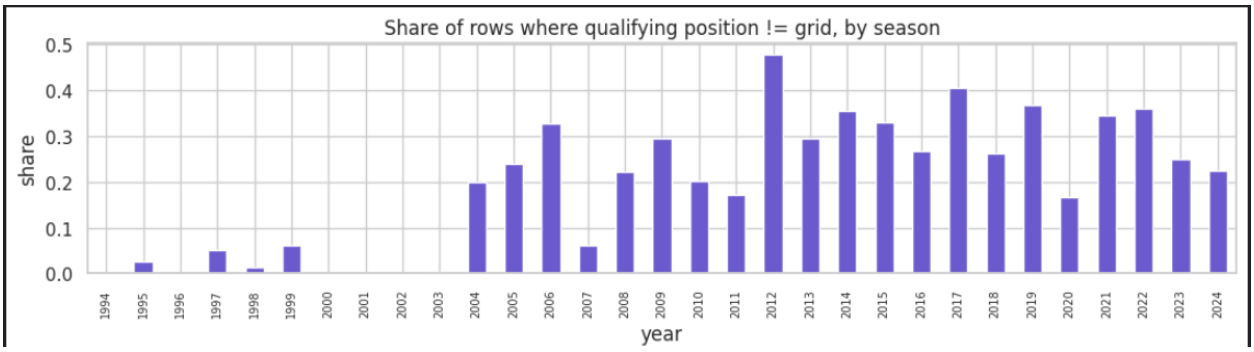


Figure 5. Share of rows where qualifying position differs from grid, by season.

- Podium rate falls from about 65% at pole to under 10% beyond P8, which supports `grid_clean` as a feature. This is association, not causation. *(Fig. 2, right panel)*

7. Leakage control and hand-off

Every column was labelled as key, target, pre-race, audit-only, filter flag or post-race. Two helper columns, `was_duplicated` and `n_rows_raw`, count as leaks, because a shared drive is only known after the race. The hand-off file `model_base_prerace.csv` has 24,724 rows × 19 columns, with keys, target and pre-race fields only. The audit file with leakage columns is not for modelling.

8. Known limitations

1. Qualifying data is absent before 1994 and patchy in 1996-2002, so qualifying features are reliable only from 2003.
2. The `laps == 0` non-starter test is a post-race proxy for 13 failure rows since 2010. None of them is a podium, and they are 0.05% of the table.
3. Shared-drive years (1950-56) have races with more than three podium rows.
4. Three unexplained qualifying anomalies: the 1999 San Marino GP swap (Villeneuve and Trulli, 5th and 14th), the 1997 British GP move (Fontana, 22nd to 14th), and the corrupted Q1 time in the 1995 Japanese GP.
5. In 2021-22 sprint weekends, `grid` is the sprint result and not the qualifying order.

Part 2: Data Analysis, Feature Engineering and Modelling

F1 Podium Prediction, Milestone 1

1. Scope and approach

This part starts from the cleaned modelling base built in Part 1 (24,724 rows, one per driver per race, 1950–2024 without the Indianapolis 500 and non-starters). It answers the three data-engineering questions with multi-table joins, then builds the leakage-safe modelling table: pre-race and past-race features, the feature selection with its evidence, the train/validation/test split, and the encoding and preprocessing pipeline. Figure numbering continues from Part 1.

2. Q1: Front-row conversion by circuit, before and after 2014

- **Joins:** `results` → `races` (year, circuit) → `circuits` (name), plus `qualifying` (qualifying position, 1:1 on raceId + driverId).
- **Grain:** one row per driver per race. A front-row start is `grid ∈ {1, 2}` (where the car actually started); a front-row qualifier is `quali_position ∈ {1, 2}`.
- **Window and missingness:** qualifying coverage is sustained only from 2003 (Part 1, Fig. 4), so the comparison is 2003–2013 vs 2014–2024. A circuit is ranked only with ≥ 5 races (≥ 10 front-row starts) in a period.
- **Cutoff:** grid and qualifying are both pre-race; only the label is post-race.

Findings

- **2014–2024:** Suzuka (89%), Yas Marina (86%) and Monaco (85%) convert best; the Hungaroring (59%), Mexico City (61%) and Baku (63%) convert worst. **2003–2013:** Magny-Cours (92%, 6 races), Istanbul (79%), and Monza and Monaco (77%) led.
- **The ranking largely changes.** Across the 14 circuits ranked in both periods, the Spearman rank correlation is 0.30 ($p = 0.29$). Conversion rose almost everywhere, from 68% to 76% of front-row starts overall. Yas Marina (+26 points), Interlagos (+20), Montreal (+18) and Suzuka (+17) rose most.
- **Qualifying vs grid:** 36 of 854 front-row qualifiers (4.2%) did not start from the front row, mainly because of grid penalties, and reached the podium only 31% of the time, against 73% for those who kept the slot. Defining the front row by qualifying changes the rate in 13 of 40 circuit-periods, by 1.8 points on average.
- **Uncertainty:** each circuit-period has 10–24 starts, so the 95% intervals are ±15–25 points. Small differences between circuits are not reliable.

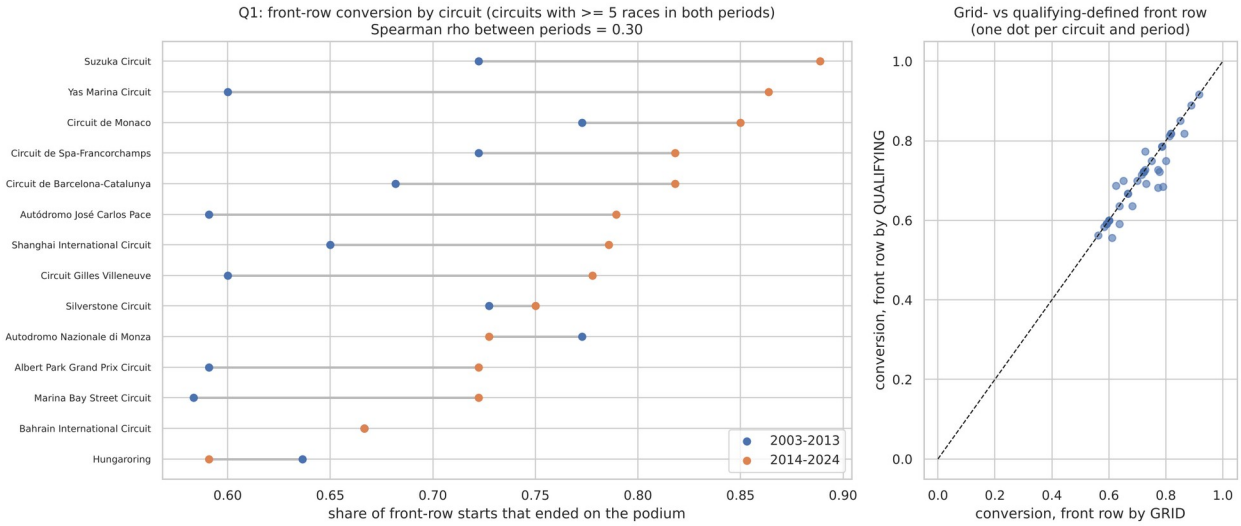


Figure 6. Front-row → podium conversion per circuit, 2003–2013 vs 2014–2024 (left), and grid- vs qualifying-defined front row (right).

3. Q2: Home-race podium rate after controlling for grid

- **Joins:** `results` → `drivers` (nationality) and `results` → `races` → `circuits` (country).
- **Nationality map:** a documented nationality → country map aligns the vocabularies ("British" → UK, "Monegasque" → Monaco). Choices: American and American-Italian → USA (the data spells it "USA" and "United States"); Argentine-Italian → Argentina; East German → Germany. Nationalities whose country never hosted a race have no home race.
- **Control for starting position:** home and away rows are compared within grid bands (P1, P2, P3, P4–6, P7–10, P11–15, P16+) and pooled with the Cochran–Mantel–Haenszel test and the Mantel–Haenszel common odds ratio. The 72 pit-lane starts (grid unknown) are excluded.

Findings

- **Raw rates:** home drivers podium *less* often (11.8% vs 13.8%) because they start further back (P12.7 vs P11.7 on average).
- **After controlling for grid:** home and away rates are close in every band (Table 1). The grid-adjusted odds ratio is **0.95 (95% CI 0.80–1.13, p = 0.58)**, so there is no home advantage.
- **Robustness:** the same holds when stratifying by exact grid slot (OR 0.96), by grid band × decade (0.92), and before or after 2003 (0.98 and 0.75, neither significant). A logistic regression with grid-slot dummies gives OR 0.96.

Table 1. Podium rate at home vs away within each starting-grid band (1950–2024).

Grid band	Home rows	Home podium rate	Away rows	Away podium rate	Difference (pts)
P1	84	61.9%	1,040	64.7%	−2.8
P2	77	57.1%	1,034	56.2%	+1.0
P3	70	52.9%	1,044	45.9%	+7.0
P4–6	227	26.9%	3,114	26.1%	+0.8
P7–10	336	7.1%	4,117	9.2%	−2.1

Grid band	Home rows	Home podium rate	Away rows	Away podium rate	Difference (pts)
P11–15	415	2.4%	5,124	2.9%	−0.5
P16+	734	0.3%	7,236	0.7%	−0.5

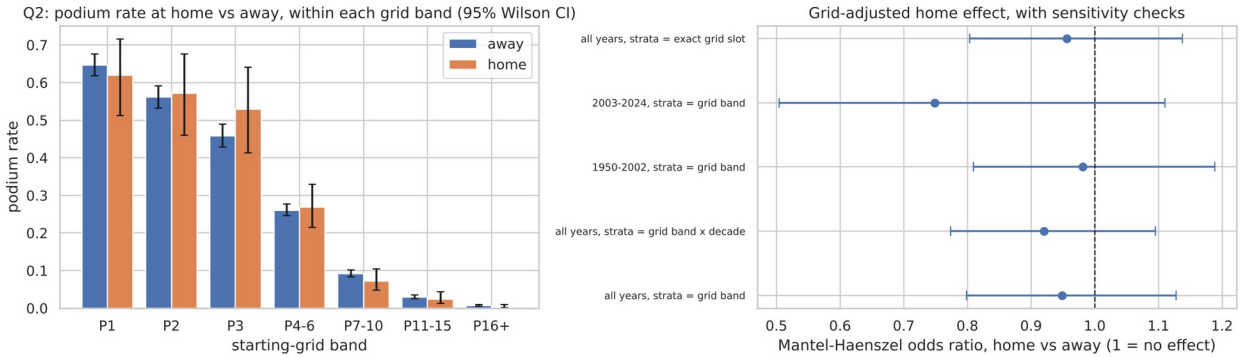


Figure 7. Home vs away podium rate per grid band with 95% Wilson intervals (left) and the grid-adjusted Mantel–Haenszel odds ratio with sensitivity checks (right).

4. Q3: Mechanical-retirement rate by constructor, 2014–2021 vs 2022–2024

- **Joins:** `results` → `status` → `races` (year) → `constructors` (name).
- **Denominator:** race starts. Rows whose status shows the car never started (Withdrew, Did not start, Illness with 0 laps) are removed, 12 rows.
- **Numerator:** every status occurring in 2014–2024 is mapped to one of five groups (Table 2), and the notebook stops if a status is unmapped. Following the brief, crashes, driver errors and regulation decisions are not mechanical. Statuses that may be a failure or a consequence of contact are kept in a separate *ambiguous* group and tested as a sensitivity check.
- **Ranking rule:** a constructor needs ≥ 30 starts in a period. Results follow `constructorId`, so renamed teams appear separately.

Table 2. Status mapping used for Q3 (race starts, 2014–2024).

Group	Statuses	Rows	Examples
finished	9	3,825	Finished, +1 Lap, +2 Laps
mechanical	36	373	Engine, Power Unit, Gearbox, Brakes, Hydraulics, ERS
accident / driver	4	305	Collision, Accident, Collision damage, Spun off
regulatory / other	3	14	Disqualified, Excluded, Illness
ambiguous	12	90	Retired (no reason), Wheel, Wheel nut, Puncture, Front wing, Debris

Findings

- **Overall:** mechanical retirements fell from 9.1% of starts (2014–2021) to 5.6% (2022–2024).

- **2014–2021:** Caterham 23.5% (34 starts), Lotus F1 17.3%, Toro Rosso 16.1%, McLaren 13.8% (the period includes its 2015–17 Honda power-unit years) and Renault 13.5% are highest; Mercedes is lowest among full-period teams at 3.8%.
- **2022–2024:** Alfa Romeo 12.5% and Alpine 11.0% are highest, then AlphaTauri 6.8%. All others are between 2.1% and 5.2%.
- **The ranking does not carry over:** for the 11 constructors ranked in both periods, the Spearman correlation is −0.50 (p = 0.12). Including the ambiguous statuses leaves the top two of each period unchanged.
- **Uncertainty:** 2022–2024 rates rest on 48–136 starts per team, so differences of about 5 points are within noise.

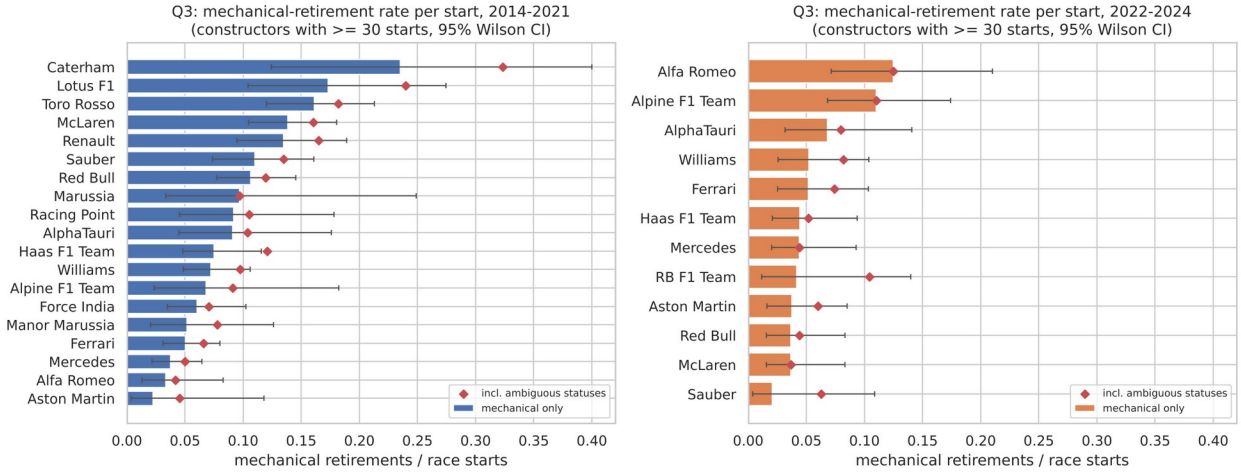


Figure 8. Mechanical-retirement rate per start by constructor (≥ 30 starts), with 95% Wilson intervals; diamonds include the ambiguous statuses.

5. Leakage-safe features

A feature for driver *d* in race *r* may use only what is fixed before the start of *r* (grid, qualifying, the calendar) or results of races that finished before *r*. History features are computed on the table sorted by race date with `shift(1)` before every rolling window, so a race never enters its own feature. The standings tables store points *after* each race, so they are re-keyed to the previous round of the same season. The history pool is the modelling base (starts only), so a non-qualification never counts as a race.

Table 3. Selected features (16) and why each is available before the race.

Group	Feature	Definition	Why it is valid at the cutoff
Grid	`grid_pct`	Grid slot ranked within the race, 0 = pole, 1 = last starter	Grid is fixed before the start
Qualifying	`quali_gap_pct`	Best qualifying lap vs session fastest, %, capped at 7% (107% rule)	Qualifying ends before the race
Qualifying	`grid_penalty`	Grid minus qualifying position (> 0 = moved back)	Penalties are applied before the start
Driver form	`drv_starts_before`	Number of earlier starts	Counts earlier races only

Group	Feature	Definition	Why it is valid at the cutoff
Driver form	`drv_podium_rate_last5`	Podium rate over the last 5 starts	shift(1) then rolling
Driver form	`drv_finish_pct_last5`	Mean relative finish over the last 5 starts (0 = winner)	shift(1) then rolling
Driver form	`drv_dnf_rate_last10`	Retirement rate over the last 10 starts	shift(1) then rolling
Driver form	`drv_career_podium_rate`	Podium rate over all earlier starts	shift(1) then expanding
Constructor form	`con_podiums_per_race_last5`	Team podiums per race, last 5 races	Team-level, shift(1) then rolling
Constructor form	`con_best_finish_pct_last5`	Team's best relative finish, last 5 races	Team-level, shift(1) then rolling
Constructor form	`con_dnf_rate_last10`	Team retirement rate, last 10 races	Team-level, shift(1) then rolling
Standings	`drv_points_share_before`	Driver points / leader's points going into the race	Standings after the previous round
Standings	`drv_wins_before`	Wins so far this season	Standings after the previous round
Standings	`con_rank_before`	Constructor championship rank going into the race	Standings after the previous round
Context	`driver_age`	Age on race day	Date of birth is known
Context	`season_progress`	(round − 1) / (rounds in season − 1)	The calendar is known in advance

- **Leakage test:** for 300 random rows, the history and standings features were recomputed from scratch on a table filtered to races strictly before that row's date. All values matched exactly. In addition, scrambling the outcome of a race leaves that race's features unchanged, and every driver's debut has no form value.
- **Sprint races** are not used as features. They exist for only 18 races (2021–2024), all outside the training years, and in 2021–22 the sprint result already set the grid, which `grid_pct` captures.
- **Identity columns are not features:** `constructorId`, `driverId`, `circuitId`, `number` and nationality are left out. Team and driver strength enter through the form and standings features instead.

6. Feature evidence and selection

All evidence uses training rows only (≤ 2019). For each of the 23 candidates the notebook reports univariate ROC-AUC, coverage, Spearman correlation with the target, and the gain in validation PR-AUC when the feature is added to a logistic regression on `grid_pct` alone (baseline PR-AUC 0.645, podium base rate 15%).

- **Starting position dominates:** qualifying position, qualifying gap and grid have AUC 0.88–0.91. The podium rate falls from about 57% in the front grid decile to below 2% beyond the midfield (Fig. 9).

- **Form and standings add the most beyond grid:** driver form and standings add +0.02 to +0.09 PR-AUC, with `drv_career_podium_rate` the largest at +0.09. Constructor form adds +0.01 to +0.02, since grid already reflects car pace. The podium rate rises steadily across their deciles, with no leak-like jump.
- **Selection rule:** keep a feature if its AUC is ≥ 0.60 or it adds ≥ 0.01 PR-AUC beyond grid; for near-duplicates (|ρ| > 0.9), keep the one with the larger gain. Dropped as weak: `is_home` (AUC 0.51, consistent with Q2), `field_size` and `grid_is_zero` (covered by the missing-value flag). Dropped as redundant: `grid_clean` and `quali_position` (ρ = 0.96–0.98 with `grid_pct`), driver rank (ρ = −0.91 with points share) and constructor points share (ρ = −0.92 with rank).
- **Least certain:** `driver_age` has a weak univariate signal (AUC 0.56) but passes on its validation gain. The validation set is small (39 races, 117 podiums), so gains of about 0.01 are within noise.

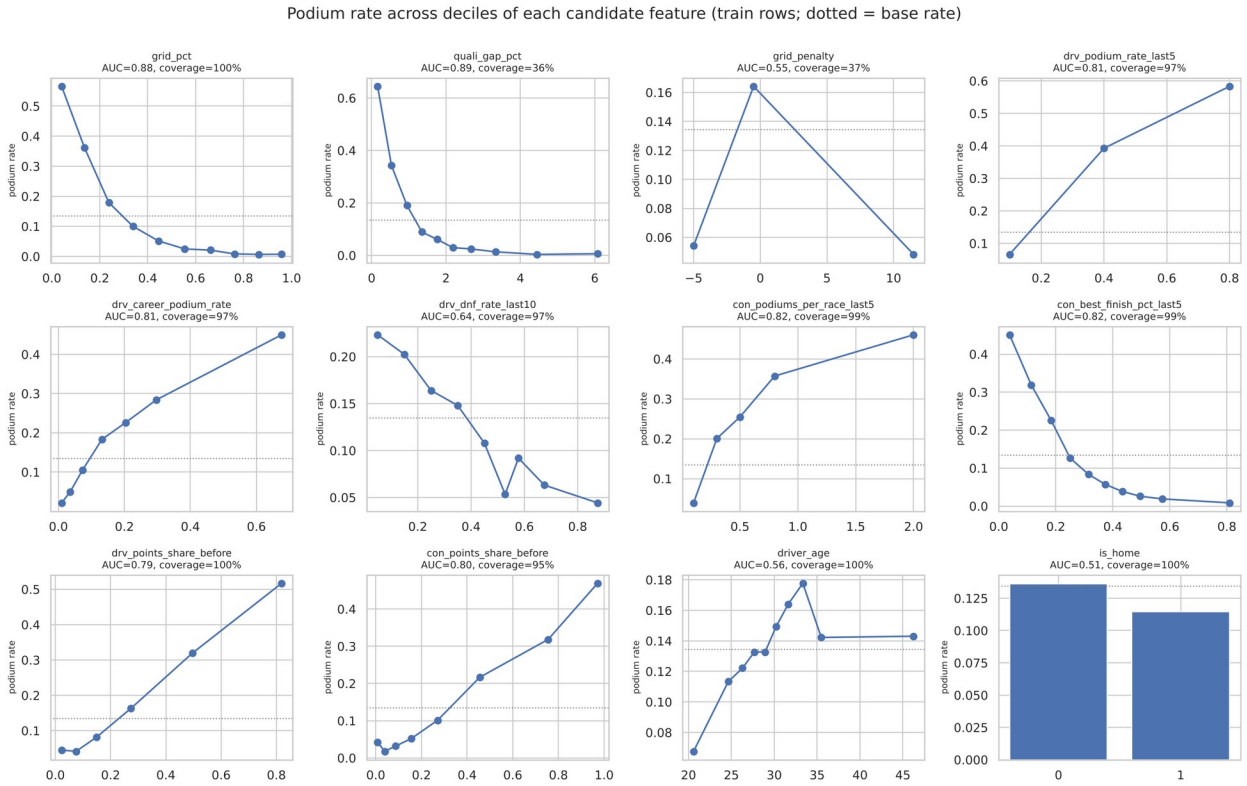


Figure 9. Podium rate across deciles of candidate features (training rows; dotted line = base rate).

7. Train / validation / test split

The split is by season: train ≤ 2019, validation 2020–2021, test 2022–2024. A random split would let the model learn team strength from future seasons, and neighbouring races share form-feature values, so they must not fall on both sides of the split. Splitting by season also keeps each race whole (its three podium places depend on each other). The test set begins with the 2022 regulation reset, a demanding test of generalisation. Assertions confirm that no race is split and that the three splits are strictly ordered in time.

Table 4. Split sizes and class balance.

Split	Seasons	Races	Rows	Podium rate
Train (full)	1950–2019	1,007	22,589	13.4%
Train (used)	2003–2019	321	6,821	14.1%
Validation	2020–2021	39	779	15.0%
Test	2022–2024	68	1,356	15.0%

- **Training window:** chosen on validation only. Start years from 1950 to 2010 score within 0.01 PR-AUC of each other; 2003 is best (PR-AUC 0.743, ROC-AUC 0.934) and 2014+ is worst. This agrees with the Part 1 finding that qualifying data is complete from 2003. Because the margin is small, 1950–2019 remains a valid alternative.
- **Unseen teams:** 16% of validation and 20% of test rows belong to constructorIds absent from training (AlphaTauri, Alpine, RB). This is a further reason not to one-hot encode team identity.

8. Encoding and preprocessing

- **Encoding:** all selected features are numeric or binary, so no one-hot encoding is needed. Target encoding was rejected: it would leak unless rebuilt out-of-fold and in time order, and the form features already carry that signal.
- **Missing values are structural:** no qualifying data before 2003, no form at a debut, no constructor standings before 1958 or at season openers, no grid for pit-lane starts. The pipeline therefore uses median imputation plus a missing-indicator column for each feature with gaps, then standard scaling. Both steps are fit on training rows only, inside an sklearn Pipeline.
- **Effect on the data:** after preprocessing every feature has mean 0 and standard deviation 1; imputation creates a spike at the median, which the indicator columns identify (Fig. 10).

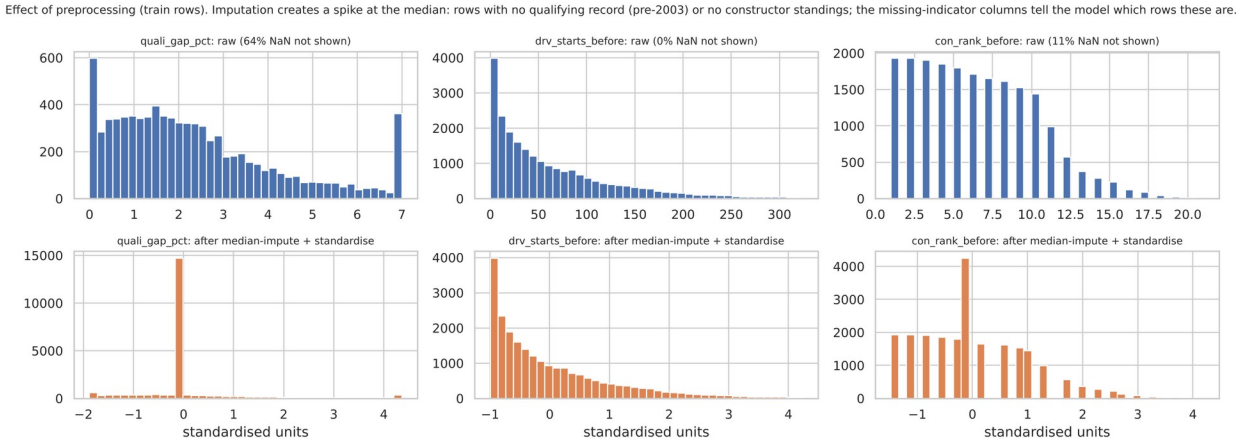


Figure 10. Three features before (top) and after (bottom) median imputation and standardisation, training rows.

Table 5. Preprocessing-order experiments (baseline logistic regression, trained 2003–2019, scored on validation; F1 threshold chosen on training predictions).

Experiment	Variant	Val. ROC-AUC	Val. PR-AUC	Val. F1
E1 order	impute → scale (used)	0.934	0.743	0.698

Experiment	Variant	Val. ROC-AUC	Val. PR-AUC	Val. F1
	scale → impute	0.933	0.740	0.698
E2 fit scope	fit on train only (used)	0.934	0.743	0.698
	fit on train + val + test (leaky)	0.934	0.743	0.692
E3 indicators	with missing indicators (used)	0.934	0.743	0.698
	without indicators	0.930	0.733	0.686

- **E1:** imputing before scaling is marginally better; it is kept because the median is then a value in real units.
- **E2:** fitting the preprocessing on all rows makes no measurable difference for logistic regression, which is insensitive to an affine rescaling. It is still a leak and is not used; models sensitive to feature ranges, such as the neural network, could be affected.
- **E3:** missing indicators give the clearest gain (+0.010 PR-AUC, +0.012 F1), which confirms that the missingness carries information.

9. Hand-off

`modeling_table.csv` holds 24,724 rows (one per raceId and driverId) with keys, year, split, the target and all 23 candidate features. `feature_spec.json` lists the 16 selected features, their groups for the ablation, and the training window. Assertions check the grain, the binary target, the row count and that no post-race column is present.

10. Predictive modelling: set-up

Three models are trained and evaluated on the modelling table from Section 9: two statistical models (**logistic regression** and a **random forest**) and a **shallow neural network** (one or two hidden layers). All three use the same rows (training window 2003–2019), the same 16 features and the same preprocessing pipeline from Section 8, so their scores are directly comparable.

- **Metrics:** ROC-AUC, PR-AUC, F1, precision and recall on the training rows, on validation (2020–2021) and on the unseen test set (2022–2024). A domain metric is added: the **top-3 hit rate**, the share of real podium places filled by the three drivers each model rates highest in a race.
- **Primary metric: PR-AUC.** Only about 15% of rows are podiums. PR-AUC measures how well podium finishers are ranked above everyone else and is not inflated by the many easy negatives at the back of the grid, which push ROC-AUC above 0.9 even for a simple grid rule.
- **Secondary metric: F1** at a fixed threshold. PR-AUC judges the ranking; F1 judges the yes/no decision the model would actually make, balancing missed podiums against false alarms.
- **Choosing settings without validation or test:** the validation set has only 39 races, so one race moves PR-AUC by about 0.01. Hyperparameters are therefore chosen by **rolling season cross-validation** inside the training window: for each season from 2014 to 2019, the model is trained on 2003 up to the previous season and predicts that season (six folds of 376–462 rows). The setting with the best mean PR-AUC is kept.
- **Decision threshold:** the probability cut-off that maximises F1 on the six folds' out-of-fold predictions. It is frozen and applied unchanged to validation and test. It is not taken from in-

sample training predictions, because a random forest predicts its own training rows almost perfectly and would get a threshold far too high for new races.

- **Final models** are fitted once on 2003–2019. Validation is used to compare them; the test set is used once, at the end. The recommended model is fixed in advance as the one with the best validation PR-AUC, preferring the simpler model when scores are within noise.

11. Reference points

- **A rule with no model:** “the drivers starting P1–P3 finish on the podium”. It is not one of the required models, but every model must beat it to justify its complexity. On validation it reaches PR-AUC 0.639 and F1 0.692 (the same rule with qualifying position: 0.654 and 0.692).
- **Why logistic regression should be a strong baseline:** it assumes each feature shifts the log-odds of a podium, log(p / (1 − p)), by a fixed amount per unit. For the dominant feature this holds closely: podium log-odds fall almost linearly with grid slot, by about 0.31 per place (Fig. 11), with only mild extra curvature at the front of the grid. If the main relationship is already close to linear in log-odds, a forest or a network has little extra shape to find.

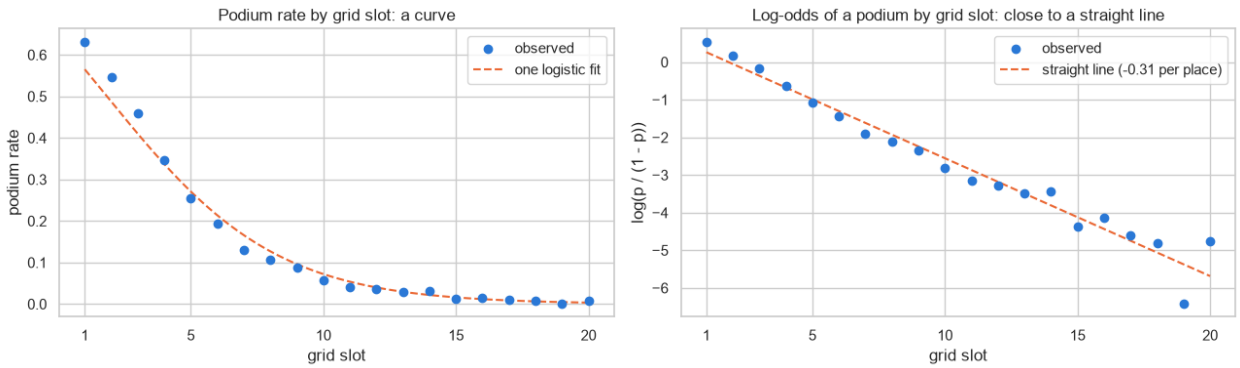


Figure 11. Podium rate (left) and podium log-odds (right) by grid slot, training seasons. One logistic fit (dashed) is a straight line in log-odds and reproduces the curved rate.

12. Models

12.1 Logistic regression (baseline)

- **How it works:** a weighted sum of the scaled features is passed through the logistic function 1 / (1 + e^−z) to give a probability. Each weight is the change in log-odds per standard deviation of a feature; training maximises the likelihood of the observed podiums. `C` sets the strength of regularisation (smaller = weights shrunk more towards zero).
- **Why the baseline:** the simplest standard model for a binary outcome, fast and transparent, and its key assumption holds for the dominant feature (Section 11).
- **Result:** `C = 1` is best (season-CV PR-AUC 0.732; only `C = 0.01` is clearly worse at 0.716). Threshold 0.39. The training PR-AUC (0.690) is below validation (0.743), so there is no overfitting; in 2020–2021 two teams took most podiums, which is easier to predict than the average 2003–2019 season.

12.2 Random forest

- **How it works:** many decision trees, each grown on a bootstrap sample of the rows and choosing each split among a random subset of features (`max_features`). The forest’s

probability is the average over 300 trees. Unlike logistic regression, it can learn curves, thresholds (e.g. the 107% rule) and interactions (e.g. grid slot × team form) without them being built as features.

- **Attempt A (default settings):** trees grown until every leaf is pure. Training PR-AUC 1.00 (the training rows are memorised), yet season-CV PR-AUC 0.746, already above logistic regression: averaging 300 different trees cancels much of each tree's noise.
- **Attempt B (tuned):** 18 settings of `min_samples_leaf` × `max_depth` × `max_features`. Best: `min_samples_leaf = 5`, no depth limit, `max_features = sqrt`, season-CV PR-AUC 0.754 (+0.008, inside the fold-to-fold spread of about 0.04). Simpler trees close the gap to the training score but slightly lower the CV score (Fig. 12). Threshold 0.29.
- **On validation** the forest scores 0.725, below the baseline (0.743), although it led in season CV.

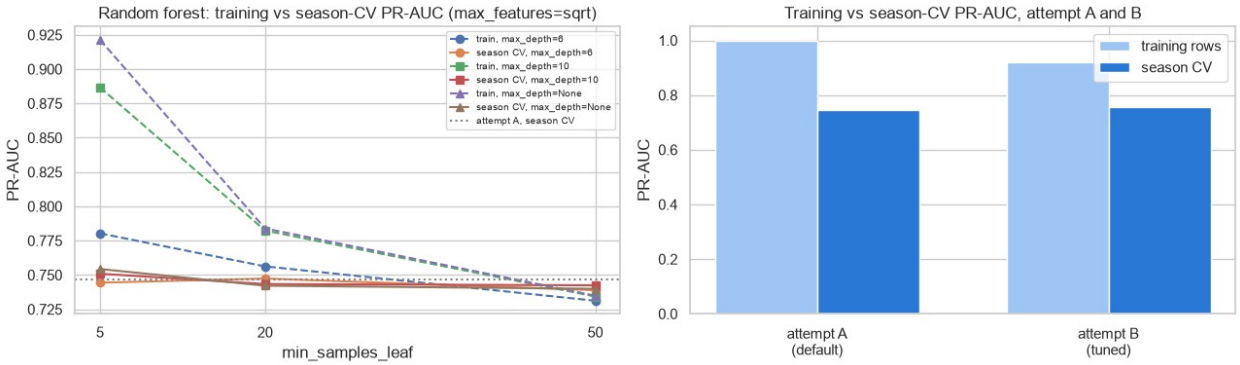


Figure 12. Random forest: training vs season-CV PR-AUC as trees may grow more detailed (left), and attempt A vs attempt B (right).

12.3 Shallow neural network

- **How it works:** scikit-learn's `MLPClassifier`. The scaled features and missing-value flags pass through one or two hidden layers of ReLU neurons; a logistic output neuron gives the probability. Weights are learned by back-propagation (Adam optimiser, log-loss); `alpha` is an L2 penalty on the weights. Scaled inputs, provided by the preprocessing pipeline, are essential for training a network.
- **Attempt A (library defaults, two layers of 32 and 16, built-in early stopping):** season-CV PR-AUC 0.719 and training PR-AUC only 0.562, i.e. underfitting. Early stopping judged on a random 10% of training rows ended training too soon.
- **Attempt B (tuned):** no built-in early stopping, 300 passes; layer sizes {8, 16, 32, 32+16} × `alpha` {0.001, 0.01, 0.1, 1}. Best: **one hidden layer of 8 neurons, `alpha = 0.01`**, season-CV PR-AUC 0.747 (+0.028). Larger two-layer networks overfit (training 0.84, CV 0.68–0.69) unless strongly penalised. With about 960 training podiums, a small network generalises best. The training loss levels off after about 50 passes (Fig. 13). Threshold 0.36.
- **Seed sensitivity:** retrained with five random seeds, the chosen network's validation PR-AUC ranges from 0.704 to 0.746 (mean 0.725). The starting weights alone move the score as much as the gap between the three models.

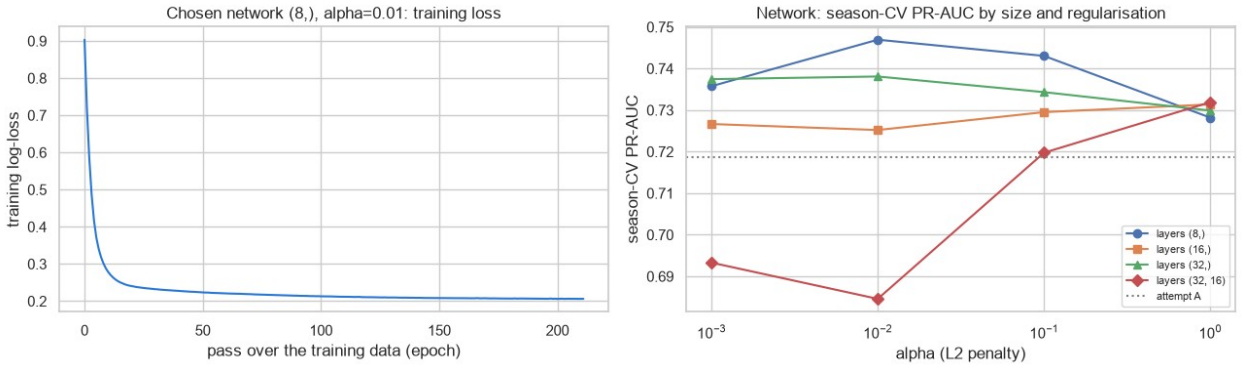


Figure 13. Shallow network: training loss of the chosen network (left) and season-CV PR-AUC by layer size and L2 penalty (right); dotted line = attempt A.

Table 6. Hyperparameter search (rolling season CV, 2014–2019) and the chosen setting per model.

Model	Search space	Chosen setting	Season-CV PR-AUC	Threshold
Logistic regression	C ∈ {0.01, 0.1, 1, 10}	C = 1	0.732	0.390
Random forest	300 trees; min_samples_leaf {5, 20, 50} × max_depth {6, 10, none} × max_features {sqrt, 0.5}	leaf 5, no depth limit, sqrt	0.754 (default 0.746)	0.286
Shallow neural network	hidden layers {8, 16, 32, 32+16} × alpha {0.001, 0.01, 0.1, 1}; 300 epochs	1 layer of 8, alpha 0.01	0.747 (attempt A 0.719)	0.355

13. Results on validation and test

Table 7 reports every metric for both rules and the three final models. Thresholds and settings come from Section 12 and were not changed after seeing validation or test.

Table 7. Model comparison: training window (2003–2019), validation (2020–2021) and unseen test set (2022–2024). F1, precision and recall at each model's season-CV threshold.

Split	Model	ROC-AUC	PR-AUC	F1	Precision	Recall	Top-3 hit
Train	Rule: grid P1–P3	.909	.626	.646	.647	.646	.647
	Rule: qualifying P1–P3	.910	.619	.637	.637	.637	.638
	Logistic regression	.932	.690	.669	.640	.700	.648
	Random forest	.985	.921	.794	.694	.928	.840
	Shallow neural network	.941	.723	.681	.634	.736	.681
Validation	Rule: grid P1–P3	.909	.639	.692	.692	.692	.692
	Rule: qualifying P1–P3	.919	.654	.692	.692	.692	.692
	Logistic regression	.934	**.743**	.692	.692	.692	.701
	Random forest	.927	.725	.694	.664	.726	.701
	Shallow neural network	.932	.717	.695	.680	.709	.692
Test	Rule: grid P1–P3	.914	.647	.649	.650	.647	.652

Split	Model	ROC-AUC	PR-AUC	F1	Precision	Recall	Top-3 hit
	Rule: qualifying P1–P3	.915	.628	.629	.631	.627	.632
	Logistic regression	.935	**.709**	.675	.664	.686	.647
	Random forest	.937	.690	**.698**	.623	.794	.662
	Shallow neural network	.935	.703	.676	.625	.735	.637

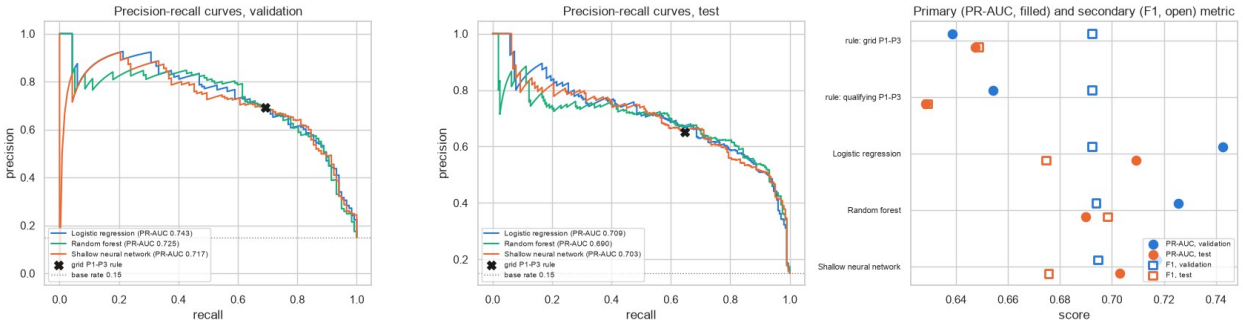


Figure 14. Precision–recall curves on validation and test (× = grid P1–P3 rule), and the primary (PR-AUC) and secondary (F1) metric per model.

Findings

- **Every model beats the grid rule on the primary metric:** test PR-AUC 0.69–0.71 against 0.65 (validation 0.72–0.74 against 0.64). The models rank podium candidates better than the grid alone, for example by rating a P2 in a dominant car above a P2 in a weak one.
- **Logistic regression is best on validation (0.743) and on test (0.709).** Season CV had ranked the forest (0.754) and the network (0.747) above it (0.732), but that order did not hold on new seasons.
- **The differences are within noise:** at most 0.02 between the three models on validation and test, smaller than the fold-to-fold spread in season CV (sd 0.03–0.05) and than the network's seed-to-seed range (0.704–0.746). No complex model reliably beats the baseline, which is consistent with the near-linear log-odds pattern in Fig. 11.
- **Secondary metric:** on validation all three models reach F1 ≈ 0.69, equal to the rule. On test the forest is highest (0.698 vs 0.675–0.676) because its lower threshold catches more podiums (recall 0.79) at lower precision (0.62).
- **Top-3 hit rate:** when each model names three drivers per race, it does no better than the grid (test: 64–66% of podium places vs 65% for the rule). The models' advantage lies in their probabilities and their ranking of the whole field, not in picking the three names.
- **Training vs validation vs test:** logistic regression and the network score about the same on training and validation (no overfitting); the forest scores 0.92 on training against 0.73 on validation. All three models lose 0.02–0.04 PR-AUC from validation to test, while the grid rule does not: after the 2022 regulation reset the form and standings features carry less information, and 20% of test rows come from teams never seen in training.
- **Recommended model: logistic regression.** It has the best validation PR-AUC (the rule fixed in advance), is confirmed on test, is the simplest and most transparent, and does not depend

on a random seed. The random forest is the alternative if catching more podiums (recall) matters more than precision.

14. Inference function

The notebook ends with `predict_podium(record)`, which predicts the podium outcome for one driver in one race from pre-race information only. It uses the recommended logistic regression, its preprocessing pipeline fitted on 2003–2019, and the decision threshold of 0.39 chosen by season cross-validation.

- **Input:** a record holding any of the 16 selected features (Table 3). Every one is known before the start.
- **Missing values:** a feature that is absent or empty is filled exactly as in training, by median imputation plus a missing-value indicator, so a record without qualifying data, a debut or a season opener still gets a prediction.
- **Checks:** fields that are not model features, such as `positionOrder` or `points`, are rejected, so post-race information cannot enter. Values outside the range seen in training are flagged.
- **Output:** podium probability, the yes/no decision at the frozen threshold, and the list of imputed or out-of-range features.
- **Demonstration (2024 Abu Dhabi Grand Prix, Lando Norris on pole, finished on the podium):** the complete record gives a podium probability of 0.648, predicted podium. A partially missing copy, without the qualifying gap, grid penalty, recent podium rate and constructor rank, gives 0.578, still predicted podium, and lists those four features as imputed. A record containing `positionOrder` is rejected. The function reproduces the pipeline's own prediction to four decimals.

15. Known limitations

- Circuit and constructor comparisons rest on small samples (10–24 front-row starts per circuit-period; 48–136 starts per team in 2022–2024), so only large differences are reliable.
- Rows from the same driver are not independent, so the Q2 confidence interval is optimistic. The nationality → country map is a modelling choice, for example for East German and dual-nationality drivers.
- The Q3 mapping is a judgement; the ambiguous group (notably "Retired" without a reason) is reported separately, and including it moves RB from 8th to 3rd in 2022–2024.
- Constructor history resets when a team is renamed (e.g. Force India → Racing Point → Aston Martin), which weakens constructor-form features for renamed teams.
- The validation set is small (39 races), so feature-selection gains and the training-window choice are sensitive to noise; the ablation measures each group's contribution after training.
- **Logistic regression** assumes additive effects that are linear in log-odds; it misses the mild extra curvature at the front of the grid and interactions unless they are built as features, and correlated features make individual weights unstable.
- **Random forest** memorises the training rows (training PR-AUC 0.92), cannot extrapolate beyond the value ranges seen in training (relevant for new teams and new rules), and its

probabilities shrink on new data, so its threshold must come from out-of-sample predictions (0.29, not 0.5).

- **Shallow neural network** results depend on the random starting weights (validation PR-AUC 0.704–0.746 across five seeds), and anything larger than a small, regularised network overfits on about 960 training podiums.
- **All models** score each driver independently, so the probabilities in a race need not add up to three podium places; they lose accuracy after the 2022 regulation change; and for the test evaluation they were not refitted on 2003–2021. Refitting on the longer window before predicting a new season would likely help slightly.
- Feature importance and associations in this report are not causal effects.

Part 3: Experiments and Explainable AI

F1 Podium Prediction, Milestone 1

1. Scope and approach

We tested what the selected model relies on. The model is the logistic regression chosen on validation PR-AUC (C = 1, 16 features, trained on 2003-2019, 6,821 rows). Every experiment below uses the same training rows and is scored on the validation set (2020-21, 779 rows, 117 podiums). PR-AUC is the primary metric, as justified in Section 10 of the previous report. Figure and table numbering continues from that report (Figures 1-14, Tables 1-7). The preprocessing-order experiments are reported there (Section 8, Table 5) and are not repeated here.

2. Feature-group ablation

We removed one feature group at a time, retrained the same logistic regression on the same rows, and scored it on validation. A larger drop means the group carries more information that the others do not. (Fig. 15)

Table 8. Feature-group ablation (groups as in Table 3; logistic regression, trained 2003-2019, scored on validation).

Group removed	Features removed	Features left	PR-AUC	Change
None (all features)	0	16	0.7426	0.0000
Grid	1	15	0.7117	-0.0309
Qualifying	2	14	0.7276	-0.0151
Standings	3	13	0.7341	-0.0085
Driver form	5	11	0.7352	-0.0074
Constructor form	3	13	0.7401	-0.0026
Context	2	14	0.7425	-0.0001

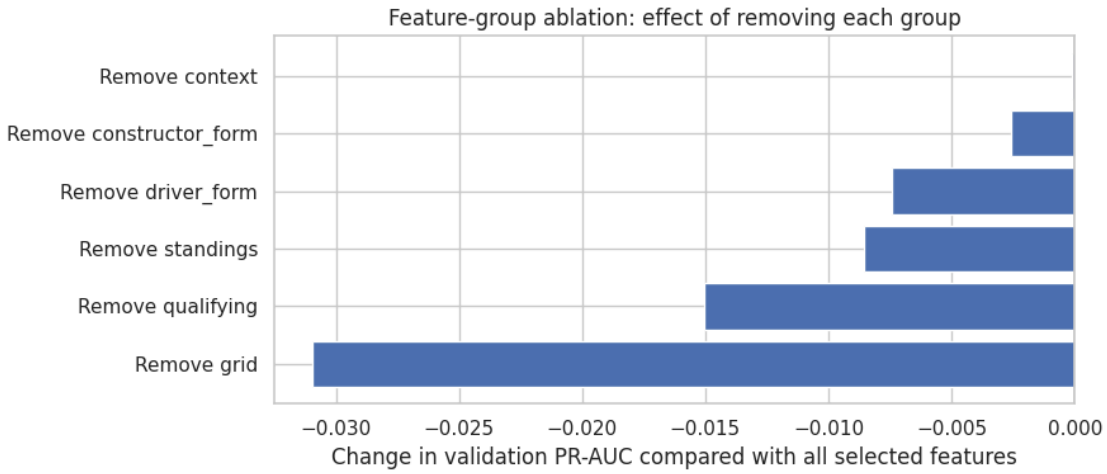


Figure 15. Change in validation PR-AUC when each group is removed.

- **Starting position matters most.** Removing grid costs 0.031 PR-AUC and removing qualifying costs 0.015. These are the only drops clearly above the noise level.
- **The drop for grid is smaller than it looks.** Without grid_pct the model still scores 0.712, above the no-model grid rule (0.639). The qualifying features (pace gap and grid penalty) still carry related starting-position information (in EDA, grid and qualifying position correlate at 0.96-0.98), which likely explains why most of the signal survives.
- **Form and standings add little on top.** Each costs 0.003-0.009, within the noise band. Removing context costs nothing (-0.0001).
- **Groups overlap**, so these drops understate each group alone and do not add up.

3. Permutation importance

We shuffled one validation feature at a time (10 repeats) in the fitted pipeline and measured the fall in PR-AUC. The model is not retrained. (Fig. 16)

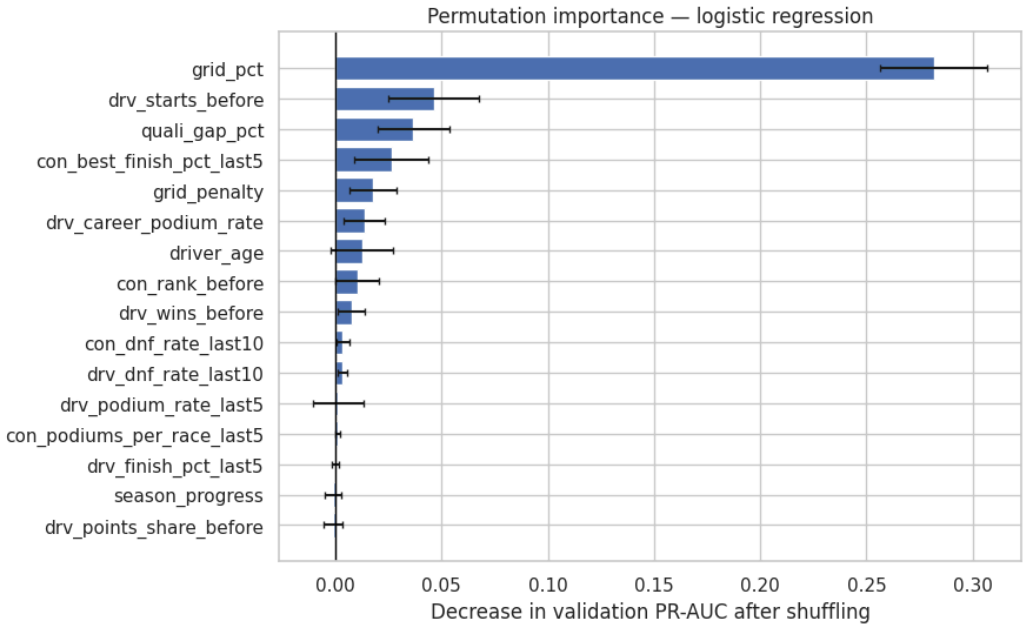


Figure 16. Permutation importance of the logistic regression (mean drop in validation PR-AUC, error bars = sd over 10 repeats).

- **grid_pct dominates:** shuffling it costs 0.282 PR-AUC, six times the next feature (drv_starts_before, 0.046). Then come quali_gap_pct (0.037), con_best_finish_pct_last5 (0.027), grid_penalty (0.018), drv_career_podium_rate (0.014) and driver_age (0.013).
- **The ranking below the top four is not reliable.** For driver_age, drv_podium_rate_last5 and con_rank_before, the sd is as large as the mean. season_progress, drv_points_share_before and drv_finish_pct_last5 sit at or below zero.
- **Low scores do not mean useless features.** The driver-form and standings features are correlated, so when one is shuffled, the others cover for it and the credit is shared.
- **Permutation and ablation answer different questions.** Shuffling grid_pct breaks something the model leans on and removes 0.282. Retraining without it lets the qualifying features take over and costs only 0.031.

4. Global SHAP

We used SHAP's linear explainer on the fitted logistic regression, in the transformed feature space, with a 100-row random training background. Values are in log-odds, not probability. Because the notebook explains the first 250 validation rows, which are date-ordered, this sample covers roughly the first twelve races of 2020, not all of validation. (Fig. 17)

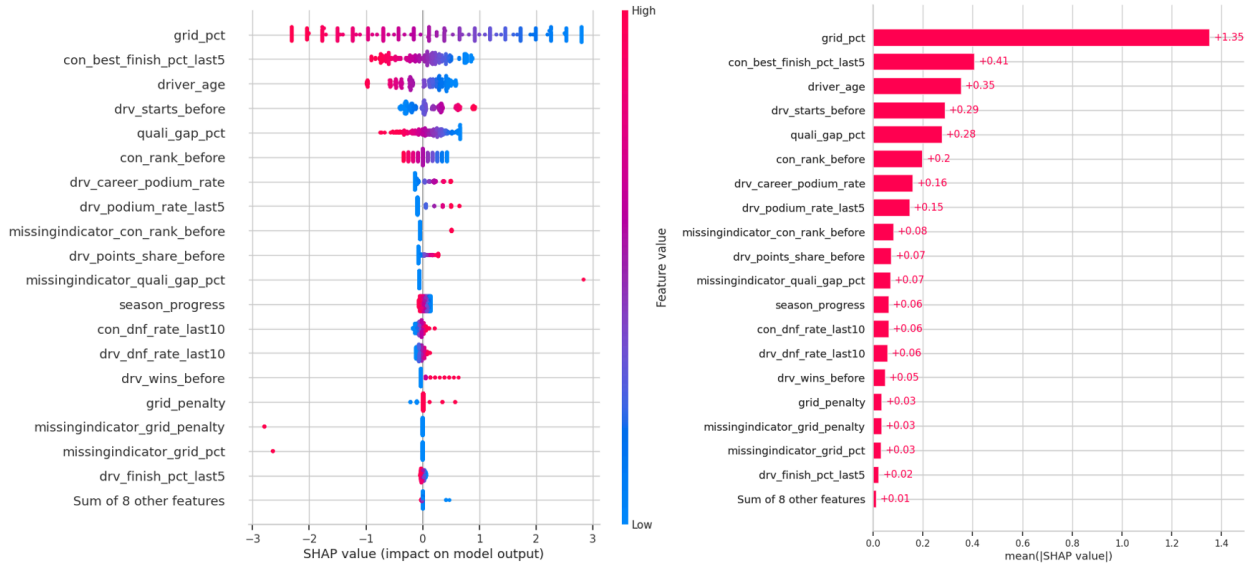


Figure 17. Global SHAP. Left: beeswarm (colour = feature value). Right: mean absolute SHAP value per feature.

- **grid_pct is first by a wide margin** (mean |SHAP| 1.35, over three times the next). Front-of-grid values (blue) push the score up, back-of-grid values push it down.
- **Next come con_best_finish_pct_last5 (0.41), driver_age (0.35), drv_starts_before (0.29), quali_gap_pct (0.28) and con_rank_before (0.20).** Better recent constructor results, more career starts and a smaller qualifying gap all push towards a podium.
- **driver_age pushes down at high values**, which looks odd next to the weak univariate signal in EDA. The vertical stripes in the beeswarm suggest that in this sample, age partly acts as a driver identifier. We did not test this.
- **Missing-indicator columns are small** (0.03-0.08) and concentrate on a few rows, such as the one record with no qualifying gap.
- **Agreement with permutation importance:** both put grid_pct first by a wide margin and share most of the top features. The middle order differs (drv_starts_before is 2nd by permutation and 4th by SHAP, driver_age is 3rd by SHAP and 7th by permutation). The two measure different things (how far a feature moves the output vs how much accuracy depends on it) and use different row samples.

5. Local SHAP: one prediction

We explained the validation podium with the highest predicted probability: raceId 1073, driverId 830 (2021), actual podium, predicted probability 0.922. (Fig. 18)

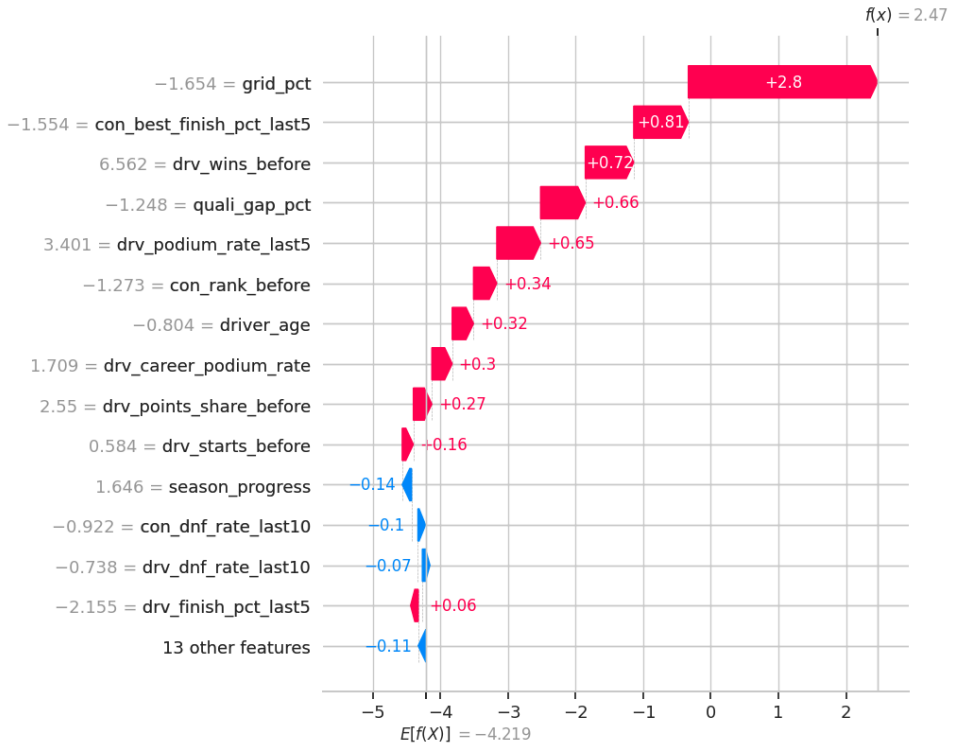


Figure 18. Local SHAP waterfall. Feature values are standardised. The baseline is E[f(X)] = -4.22 log-odds, and the prediction is f(x) = 2.47.

- **The prediction sits 6.7 log-odds above the baseline.** The baseline is the average log-odds of the background rows (about 1.4% as a probability), pulled well below the 13% podium rate by the many back-of-grid rows.
- **grid_pct alone contributes +2.8**, about 42% of the total, because the driver starts near the front (-1.65 sd).
- **The rest comes from strong form:** best recent constructor finishing (+0.81), 6.6 sd more career wins than average (+0.72), qualifying gap (+0.66) and recent podium rate (+0.65). No feature pushes meaningfully against the prediction (the largest negative is -0.14).
- **This is the easiest case by construction.** It shows how the model combines signals when they all agree, not how it behaves on borderline or wrong predictions.

6. Known limitations

- The ablation, permutation importance and SHAP all use the validation set (39 races, 117 podiums), where differences of about 0.01 PR-AUC are within noise. Each experiment was run once, without repeated seeds or cross-validation.
- Only the logistic regression is explained. The random forest and neural network were not.
- Correlated features share credit in permutation importance and SHAP, so a low score for one feature does not show that its information is useless.
- Global SHAP covers the first 250 validation rows only (about the first twelve races of 2020).
- The local explanation is a single, easy example.

- Importance is not causation. A fast car plausibly explains both a good grid slot and a podium, so a high importance for grid_pct says nothing about whether starting position itself causes the result.